Assignment no. 4

Generative AI

GAI4UE

SS 2026

This assignment is the **inverse** of HW 3. Where pix2pixHD generated brightfield images from masks, here you will train a **U-Net** to predict the foreground mask from a brightfield image. You will use `segmentation_models_pytorch` with an EfficientNet-B0 encoder pretrained on ImageNet, then evaluate with Dice and IoU. Finally, you will probe the model's **robustness** under input noise.

Exercise 1 (4 points) Prepare BBBC010 for segmentation:

- Download `BBBC010_v2_images.zip` and `BBBC010_v1_foreground.zip`.
- Pair each well's brightfield (w2) image with its binary mask, normalise the image to 8-bit RGB, and resize everything to **512×512**.
- 80/20 train/test split (`random.seed(42)`);

Exercise 2 (8 points) Train a U-Net (EfficientNet-B0 encoder):

- Build model.
- Loss = **0.5 BCE + 0.5 Dice** (use `smp.losses.DiceLoss`).
- Optimizer: Adam (`lr` $= 10^{-4}$); LR schedule: `CosineAnnealingLR`; **30 epochs**.
- Apply Albumentations augmentations on train only (flips, `RandomRotate90`, mild affine).
- Save the checkpoint with the best validation Dice.
- Plot training/validation loss and validation Dice/IoU curves.

Exercise 3 (4 points) Evaluate on the test set:

- Load the best checkpoint and compute **per-sample Dice and IoU** on all 20 test wells.
- Report mean±std, min, max for both metrics; plot histograms.
- Show the **two best** and **two worst** samples — input, ground truth, prediction (binary), prediction (probability).

Exercise 4 (4 points) Adversarial robustness — Gaussian noise added:

- Add zero-mean Gaussian noise in input-pixel space ($[0, 1]$) at $\sigma \in \{0, 0.001, 0.002, 0.005, 0.010, 0.015, 0.020, 0.030\}$.
- At each σ, recompute IoU and Dice over the full test set; plot the mean curves.
- Visualise one representative sample at every σ (noisy input on top, prediction below).
- Discuss: at which σ does mean IoU drop below 0.5? What does this imply for deployment (e.g. how to mitigate)?

Skeleton: A starter repository with notebook templates is available at:
`https://github.com/MustafaArikan/GAI4_course` (folder `HW_4`).

Deliverables:

- 3 Jupyter notebooks exported as HTML
- 1-page report (PDF): summarise final Dice/IoU, comment on best/worst samples, and discuss the IoU-vs-σ curve, where the model breaks and what that implies for deployment (how to mitigate?).

Submission: electronically via Moodle; **Deadline: Mon, Jun 15, 2026, 23:59**